

Sales & Customer Insights Analysis

Introduction

This project focuses on analysing retail transaction data to extract meaningful insights about sales performance and customer behaviour. The analysis helps in understanding business trends and improving decision-making.

Dataset

The project uses the Istanbul Shopping Dataset, which includes transaction data from shopping malls between 2021 and 2023.

The dataset contains:

- Customer demographics (age, gender)
- Transaction details (invoice number, quantity, price)
- Product categories
- Payment methods
- Shopping mall locations

Objectives

- To analyse revenue trends over time
- To identify top-performing product categories and locations
- To understand customer demographics
- To evaluate payment preferences

Tools Used

SQL (MySQL) for data analysis

Power BI for visualization

Data Preparation

- Checked for missing values and duplicates
- Verified data types
- Created derived metric:
- $\text{Revenue} = \text{Price} \times \text{Quantity}$

Key Metrics

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value (AOV)

Analysis Performed

- Revenue by Category
- Revenue by Shopping Mall
- Monthly Revenue Trend
- Revenue by Payment Method
- Revenue by Gender
- Revenue by Age Group
- Top Customers Analysis

Dashboard Description

Sales Overview

- KPI Cards (Revenue, Orders, Customers, AOV)
- Monthly Revenue Trend
- Revenue by Category
- Revenue by Shopping Mall
- Revenue by Payment Method
- Revenue by Age Group
- Revenue by Gender

Key Insights

- A few categories generate the majority of revenue
- Sales trends vary across months, indicating seasonal patterns
- Credit cards are the most commonly used payment method
- Customers aged 25–40 contribute significantly to revenue
- High-value customers account for a large share of total sales

Conclusion

This project demonstrates the effective use of SQL and Power BI to analyse retail data and generate actionable insights. The dashboard provides a clear and interactive view of business performance, enabling better decision-making.